



AUTUMN END SEMESTER EXAMINATION-2024

3rd Semester B.Tech

AUTOMATA THEORY & FORMAL LANGUAGE

CS-21003

(For 2024 (L.E), 2023 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes

Full Marks: 50

Answer any FIVE questions.

Question paper consists of two SECTIONS i.e. A and B.

Section A is compulsory.

Attempt any Four question from Sections B.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

- (a) Find the number of states in the minimal deterministic finite automaton corresponding to the regular expression $(a + b)^*ab$.
- (b) Consider the following two regular expressions over the alphabet $\{0, 1\}$:

$$r = 0^* + 1^*$$

$$s = 01^* + 10^*$$

Find the strings of length less than or equal to 3, which are neither in r nor in s .

- (c) Which of the following language(s) is/are regular? Justify.
 - (i) $\{a^n b^n \mid n \geq 0\}$
 - (ii) $\{a^k b^k \mid 0 < k < 5\}$
 - (iii) $\{a^m b^m \mid m \geq 1\}$
 - (iv) $\{a^m b^n a^n \mid m \geq 1, n \geq 0\}$
 - (v) None of these

(d) Context-free languages are closed under

- (i) intersection, complement
- (ii) union, intersection
- (iii) union, kleene closure
- (iv) complement, kleene closure
- (v) None of these

Justify your answer.

(e) A CFG G is given with the following productions where S is the start symbol, A is a non-terminal and a and b are terminals.

$$S \rightarrow aS \mid a$$

$$A \rightarrow aAb \mid bAa \mid \epsilon$$

Find the language of the above grammar.

(f) Consider a grammar G with productions $S \rightarrow SS \mid a \mid b$. Show that the given grammar is ambiguous.

(g) Find a CFG without useless productions equivalent to the grammar G defined by:

$$S \rightarrow aS \mid A \mid C$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

$$C \rightarrow aCb$$

(h) Let L_1 be the set of all languages accepted by a PDA by final state and L_2 the set of all languages accepted by empty stack. Which of the following is true?

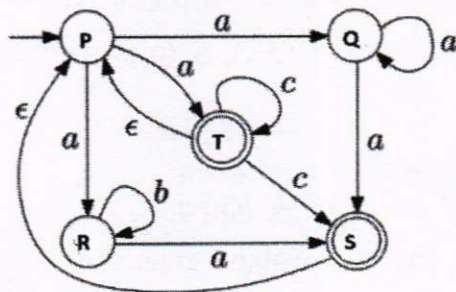
- (i) $L_1 = L_2$
- (ii) $L_1 \subset L_2$
- (iii) $L_2 \subset L_1$
- (iv) $L_2 \subseteq L_1$
- (v) None of these

(i) What is the Chomsky Hierarchy in formal language theory?

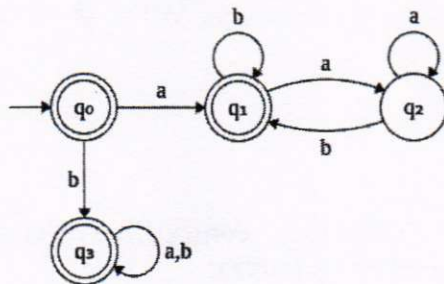
- (j) Write the example of two context-free languages whose intersection is not a context-free language.

SECTION-B

2. (a) Construct a DFA over input alphabets $\Sigma = \{a, b\}$ for language, $L = \{a^n b^m \mid n \geq 0, m \geq 0, m + n \geq 2\}$ [5]
 (b) Convert the following ϵ -NFA into its equivalent DFA. [5]



3. (a) Construct the regular expression using the state elimination method for the following DFA, eliminating the states in the order q_0, q_2, q_3, q_1 . [5]



- (b) Find a right-linear grammar for the following language: [5]

$$L = \{a^m b^n \mid m \geq 0, n > 0\}$$

Convert it into an equivalent left-linear grammar.

4. (a) Design the CFG for the following languages: [5]
 (i) $L_1 = \{w \in \{a, b\}^* : n_a(w) = n_b(w)\}$
 (ii) $L_2 = \{a^n b^m c^k : n = m \text{ or } m = k\}$

- (b) Transform the following CFG into Chomsky Normal Form (CNF): [5]

$$S \rightarrow baAB$$

$$A \rightarrow bAB \mid \epsilon$$

$$B \rightarrow BAa \mid A \mid \epsilon$$

5. (a) Construct a NPDA that accepts the language over $\{a, b, c\}$ [5]

$$L = \{a^m b^n c^p : m \geq 0, n \geq 0, p \geq 0, 2m = n + p\}$$

- (b) Convert the following CFG, G to an NPDA, N such that $L(G) = L(N)$. [5]

$$S \rightarrow aB \mid bA$$

$$A \rightarrow aS \mid b$$

$$B \rightarrow bS \mid a$$

Find the language of the above grammar.

6. (a) Construct a Turing Machine for the following language: [5]

$$L = \{wcw \mid w \in \{a, b\}^*\}$$

- (b) Write the sequence of Instantaneous Descriptions (IDs) for the following input strings to the above designed TM. [5]

i) bbacaba

ii) abbcabb

7. (a) Convert the following context-free grammar to Greibach Normal Form (GNF): [5]

$$S \rightarrow AA \mid b$$

$$A \rightarrow aS \mid \epsilon$$

- (b) State Pumping Lemma for context-free languages. Using this prove that the following language is not a context-free language. [5]

$$L = \{a^n : n \text{ is a perfect square}\}$$
